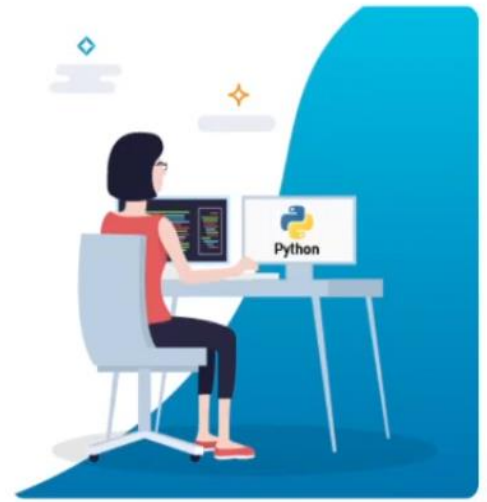


Prerequisites

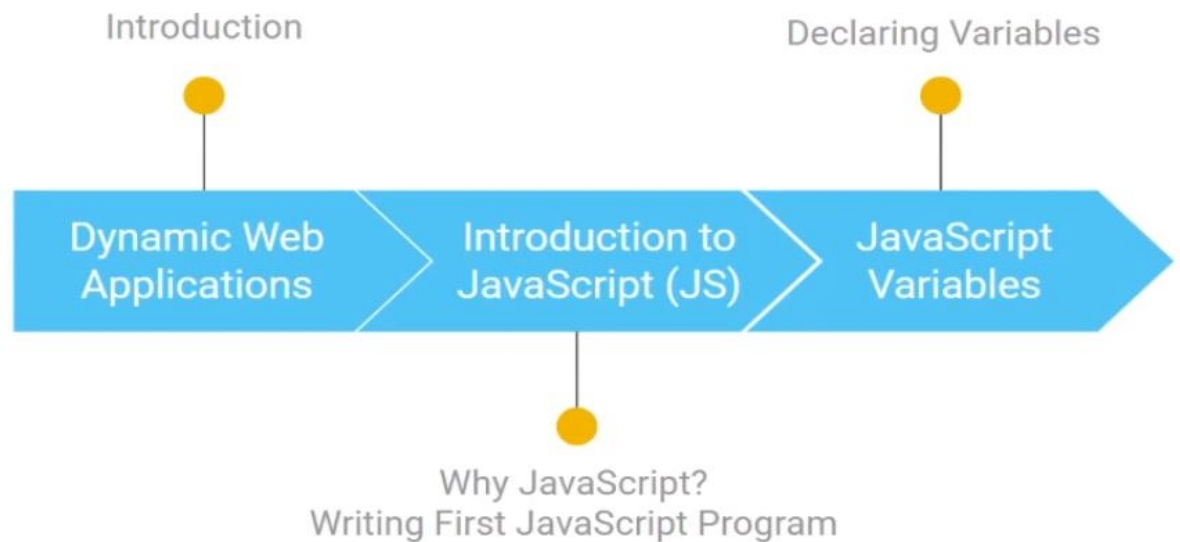


Tech Foundations

Programming Foundations

Agenda

Introduction to Dynamic Web Applications



Dynamic Web Applications



The image displays a collection of six video thumbnails arranged in two rows of three. The top row includes a video titled '2D DRAWING' with a duration of 32:24, a video of an ancient Egyptian temple with a duration of 19:56, and a video titled 'WILD COAST' with a duration of 4:46. The bottom row features a video titled 'Moving India Part-2' with a duration of 1:11:67, a video titled 'NETFLIX OUR PLANET FULL EPISODE' with a duration of 49:28, and a video titled 'American Academy' with a duration of 9:01. A YouTube logo is visible in the top left corner of the grid.

Commentary	Scorecard	Highlights	Full Commentary	Live Blog	Match Facts	News
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IND 113/2 (22.2) CRR: 5.06						
India opt to bat						
Batsman	R	B	4s	6s	SR	Key Stats
Virat Kohli *	39	50	4	0	78.00	Partnership: 31(41)
Shreyas Iyer	19	20	2	0	95.00	Last Wkt: Shubman Gill bow'd Agar 33(38) - 82% in 15.4 ov
Bowler	O	M	R	W	ECO	Last 5 overs: 26 runs, 0 wickets Took: India (Batting)
Adam Zampa *	4.2	0	20	0	4.60	
Ashton Agar	4	0	16	1	4.00	

Recent: .. 0 1 0 1 | 1 0 1 1 1 0 1 1 1

[Have Your Say](#)

22.2 Zampa to Shreyas Iyer, 1 run, to sweeper cover

22.1 Zampa to Kohli, 1 run, tossed up outside off, Kohli gets a good stride forward and whips it away down to long-on

22	Runs Scored 4 1 0 1 1 1 1 0	Score after 22 overs IND 111-2	Shreyas Iyer Virat Kohli	18(19) 38(49)	Ashton Agar 4-5-16-1
-----------	--------------------------------	--	-----------------------------	------------------	-------------------------

21.6 Agar to Shreyas Iyer, no run, goes fuller on middle and off, Iyer hangs on the back foot and punches it back towards Agar

21.5 Agar to Kohli, 1 run, shortish around off, Kohli camps back and dabs it late to short third man. Tight single taken

21.4 Agar to Shreyas Iyer, 1 run, makes a hint of room early by backing away and punches this to sweeper cover

all W: If India play like their traditional way, they will end up scoring near 310-315 and that's not

Real-Time

Live

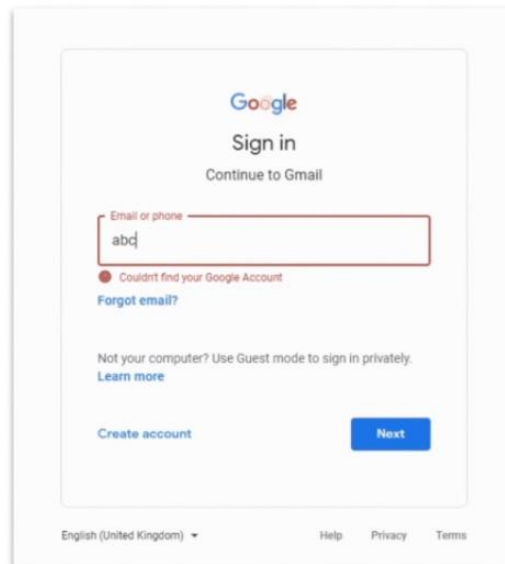
Traffic Updates



User Interactivity

Form

Data Validations



The image shows a Google Sign-in page. At the top is the Google logo, followed by 'Sign in' and 'Continue to Gmail'. Below this is a text input field labeled 'Email or phone' containing the text 'abc'. To the right of the input field is a red error message: 'Couldn't find your Google Account'. Below the error message is a link 'Forgot email?'. Further down, there is a link 'Not your computer? Use Guest mode to sign in privately. Learn more'. At the bottom left is a link 'Create account' and at the bottom right is a blue 'Next' button. The footer contains 'English (United Kingdom)', 'Help', 'Privacy', and 'Terms'.

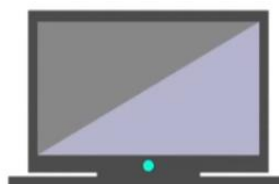


How can we build Dynamic Web Applications?

³⁷⁷¹⁴³Building Process

Building Dynamic Web Applications
mainly involves:

- Client Server Communication
- Manipulating HTML and CSS
- Writing Application Logic



Ways to achieve

- Using JavaScript
- Using Web Assembly



Why JavaScript?

Understood by Major Browsers

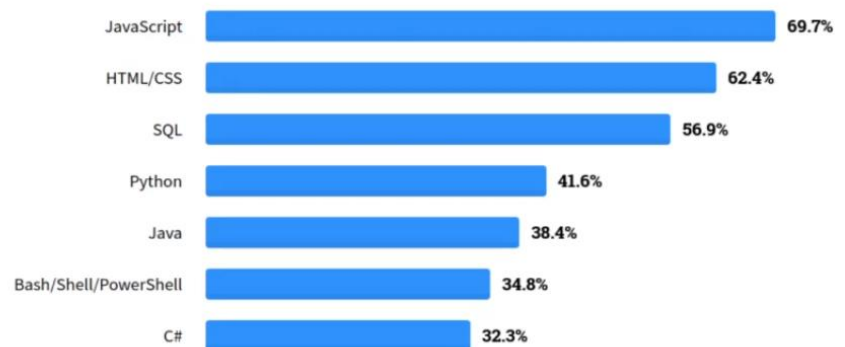
JavaScript is the most popular programming language which is adopted and **understood** by all the major Browsers



Why JavaScript?

Most Popular Technology

~69.7% developers
use JavaScript



Why JavaScript?

Companies Using JavaScript



...many more

JavaScript (JS)

HTML provides the basic **structure** for web pages.

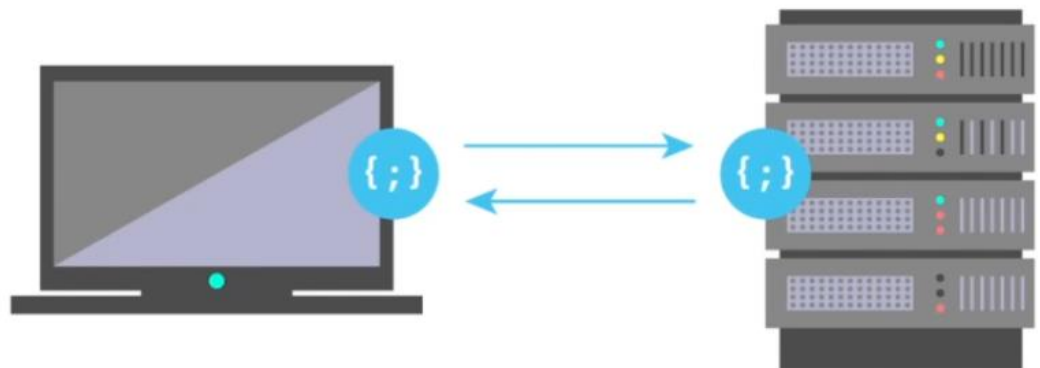
CSS is used for **styling**.

JS is used primarily by web browsers to create a **dynamic** and **interactive** experience for the users.

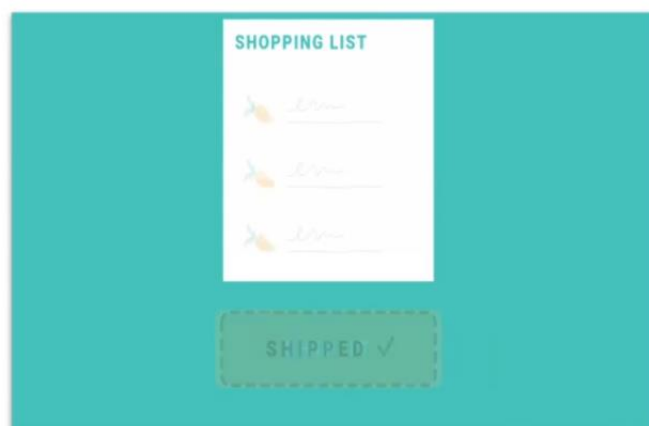


JavaScript

Can be used in **both** client-side and server-side.



Button Animations



JavaScript Applications

Live Dashboards



Programming with JavaScript

Programming with JavaScript

Your First Program

Displaying a Message

Writing a program to display "Hello World!"

```
console.log("Hello World!")
```

Hello World!

JavaScript Variables

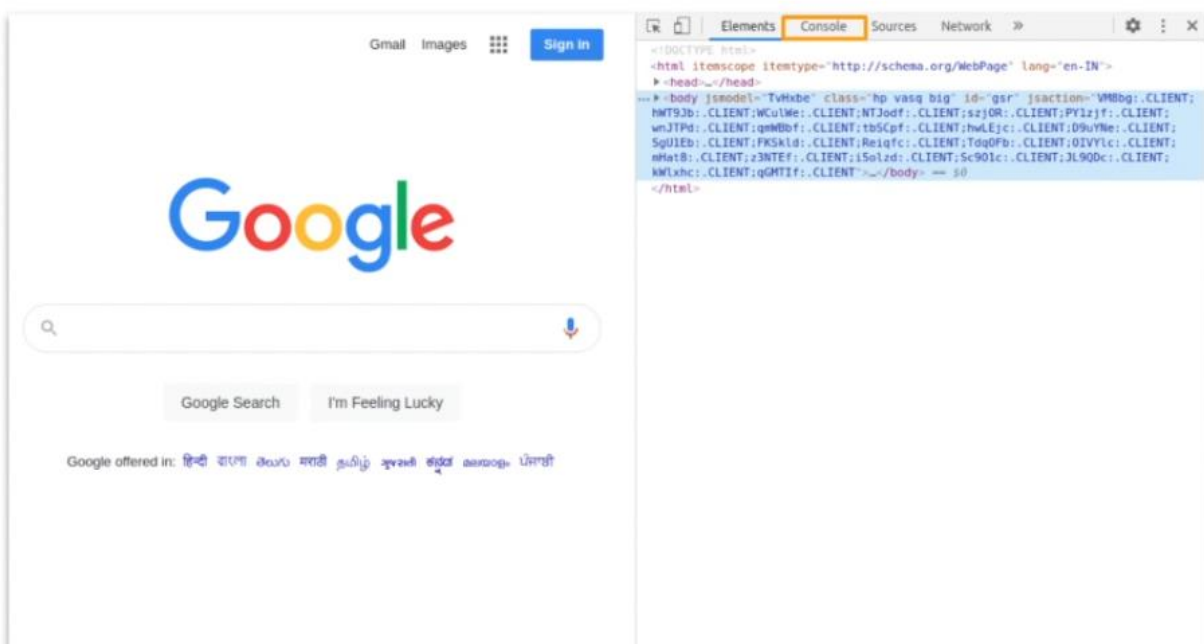
Possible Mistakes

Code	Possible Mistake
<code>consolelog("Hello World!");</code>	Missing . between
<code>Console.log("Hello World!");</code>	Uppercase 'C' in console
<code>console.log(Hello World!);</code>	Missing quotes
<code>console.log("Hello World!"</code>	Missing Parentheses

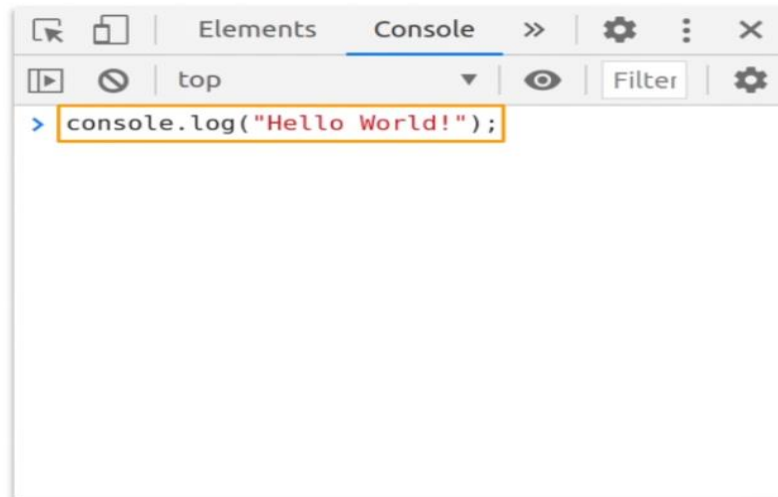
Right Click on Webpage and Click on Inspect



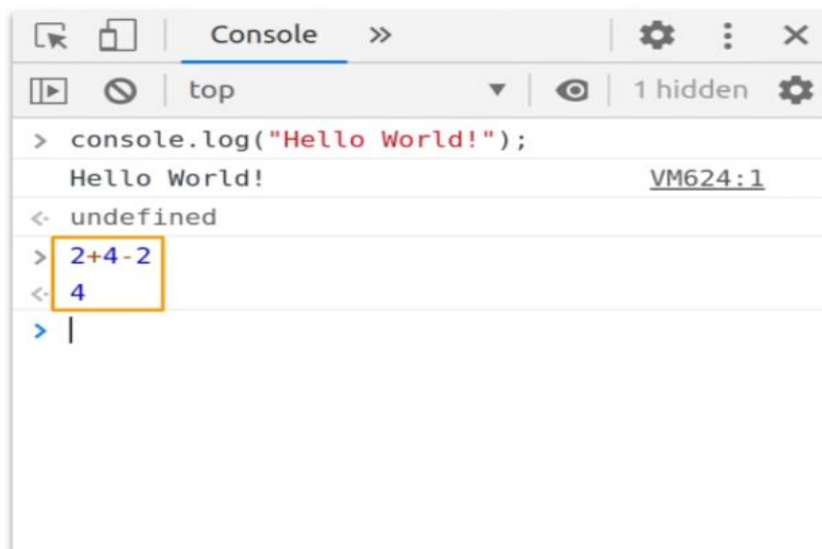
Click on Console



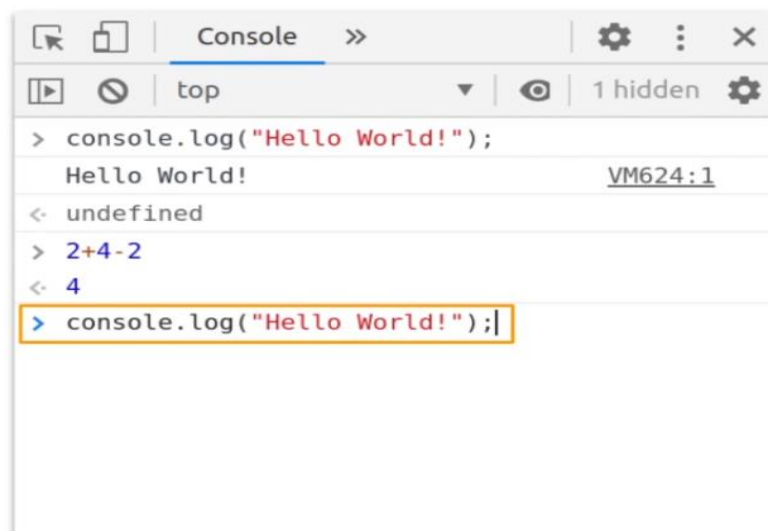
Write Your Code



Using as Calculator



Up/Down Arrow Key Press



JavaScript Variables



JavaScript Variables

Variables

Variables are like containers
for storing values



JavaScript Variables

Declaring Variables

We can create a variable using
`let` keyword

```
let message;
```

Code

Assigning Value

We can put data into a variable using **assignment operator** (=)

```
let message = 'Hello Rahul';
```

Code

```
let message;  
message = 'Hello Rahul';
```

Code

Declaring Variables

Printing a variable without assigning a value will give **undefined** as output

```
let message;  
console.log(message);
```

Code

undefined

Output

Key Takeaways

- Dynamic Web Applications
- Introduction to JavaScript
 - Why JavaScript?
 - Writing First JavaScript Program
- JavaScript Variables
 - Declaring Variables
 - Assigning Values

